



Review

Removal of cyanobacterial and algal cells from water by ultrasonic waves – A review



Mohammad Hadi Dehghani

Tehran University of Medical Sciences, School of Public Health, Department of Environmental Health Engineering, Tehran, Islamic Republic of Iran
 Tehran University of Medical Sciences, Institute for Environmental Research, Center for Solid Waste Research, Tehran, Islamic Republic of Iran

ARTICLE INFO

Article history:

Received 15 September 2015

Accepted 2 August 2016

Available online 3 August 2016

Keywords:

Sonodegradation

Cyanobacteria

Algae

Cyanotoxin

Water

ABSTRACT



The importance of cyanobacteria and algae cells in water is attributed to the diverse effects they have on an aquatic environment. The decomposition of certain compounds from live and dead algae gives water a taste and smell. It is proposed in this study that ultrasonic waves may provide an environmentally friendly method to inhibit the growth rate of cyanobacteria and algae in water. The aim of this study was to review ultrasonication for the removal of cyanobacterial and algal cells and cyanotoxins from water. In this article, effects of the operational parameters of exposure time, frequency, intensity, power level and these parameters influence on cyanobacterial and algal bloom (removal percentage, content of chlorophyll *a* etc.) are reviewed.

© 2016 Elsevier B.V. All rights reserved.

Contents

1. Introduction	1109
2. Negative significance of cyanobacterial and algal bloom in water	1110
3. Studies on sonodegradation of cyanobacterial and algal cells	1110
4. Sonochemical degradation of cyanotoxins from water	1112
5. Conclusions	1113
References	1113

1. Introduction



Ultrasonic sound is the part of the sound wave spectrum that ranges from 20 KHz to 10 MHz. The ultrasonic range from 20 KHz to 1 MHz is used in chemistry applications of ultrasonic waves. Chemical sound is the use of ultrasonic waves to accelerate chemical reactions and processes. Ultrasonic waves are produced for industrial use by vibrations from piezoelectricity and magnetostrictive. Based on the effect of **piezoelectricity**, some crystal matter is changed mechanically and in thereby forms one electrical square. In this situation ultrasonic waves are produced [1–6].

Ultrasonic waves cause a pressure gradient, cavitation and vibration bubbles in an environment that can have mechanical, thermal and chemical effects in an environment. All solutions contain a significant amount of gas bubbles. The effect of a mechanical quake causes these bubbles to reach a certain diameter at specific wavelengths, ultrasonic waves (6 μm in diameter, 1 MHz frequency) and cause characteristics in their resonance in such a way that amplitude oscillations will be bigger. A big swing within range solutions can cause changes to biological tissue by cell wall rupture and the rupture of large molecules. However, due to severe fluctuations and high pressure and variations of gas inside bubbles, a phenomenon similar to gas ionization produces free radicals and causes a higher density of radicals in and around the water. These radicals can produce various chemical compounds. This type of cavitation is called stable cavitation. **Another type of cavitation is termed transition cavitation that only occurs at a very high intensity of ultrasound energy [7–24].**

E-mail address: hdehghani@tums.ac.ir.

The aim of this study was to review the impact of ultrasonic irradiation at different exposure time, frequency, power or intensity and removal percentage of cyanobacterial, algal cells and toxins in water.

2. Negative significance of cyanobacterial and algal bloom in water



The bloom of cyanobacteria or algae in water can change its quality and cause problems with its treatment. Cyanobacteria and algae bloom make worse physical water quality indicators such as: color, taste, odor, turbidity. During treatment processes of water plenty of blue-green or green algae bloom is consumed higher doses of coagulant. Cyanobacterial or algal biomass can cause clog up of filters and cause also technological losses. Moreover, cyanobacteria can produce cyanotoxins (hepatotoxins, neurotoxins, cytotoxins) which can be released during water treatment processes. There are many reports about fatal cases of poisoning of humans and animals due to presence cyanotoxins in drinking water [25–36].

The removal of harmful algae from water treatment plants and drinking water is difficult because of its small size and low specific gravity. To control excess growth of algal bloom, research has been done on using technologies such as ultraviolet, ozone, chlorine dioxide, chlorine, potassium permanganate or preoxidant, coagulation, flotation, filtration and other advanced technologies. Some of these technologies are complex, costly and may cause secondary contamination [37–41].

3. Studies on sonodegradation of cyanobacterial and algal cells



In recent years, the application of ultrasonic waves as a novel technology for the removal of harmful pollutants and toxins from water and wastewater has attracted considerable attention. The impact of ultrasonic waves on cells is variable according to the power of sonication, sound intensity, time and frequency, as well as the physiological state and structure of the microbiological cell [2,4,7–11,16,19,20,21].

Ahn et al. suggested that at a frequency of 22 kHz (0.63 W cm^{-3}), the cyanobacterial concentration in control samples increased to 91%. This study showed that the *M. aeruginosa* cell density and chlorophyll *a* rapidly decreased after 3 days of ultrasonication [1,2].

The impact of ultrasonication (8 W, 16 W and 18 W), 20 kHz at exposure time of 5 s was tested on the growth rate and chlorophyll *a* concentration of *S. maxima* by Al-Hamdani et al. (1998). The study showed that a lower dose of ultrasonic waves (8 W) resulted in a 5% increased growth rate on the first 7 days of the experiments. Also, chlorophyll *a* concentration of *S. maxima* after sonication (8 W and 20 kHz) increased from day 7 to 14 [3].

Sonoreactor was applied in a study by Broekman et al. In this study *S. capricornutum* cells were exposed to ultrasonic waves (1.5 MHz , 10 W cm^{-3}). This experiment indicated that ultrasonication can cause lysed damage in algal cells [12].

Walsby (1994) suggested that ultrasonic waves with low frequency and high power intensity were not suitable for reducing growth rate and vacuole damage [31].

Nakano et al. observed that using ultrasonic waves at 20 KHz and water jet circulation improved the reduction of cyanobacterial growth [42]. Other surveys on *A. flos-aquae* have shown that a low dose (50 W), low frequency (20 kHz) and a short sonication time (5 min) enhanced cell growth [43,44].

Thomas et al. demonstrated that cyanobacteria and algae growth was affected by sonication time, frequency and power. The research demonstrated that growth rates of *A. flos-aquae* increased with ultrasonic waves and growth rates of *S. capricornutum* decreased after ultrasonication [45].

Purcell et al. (2009) studied ultrasonication of filamentous cyanobacteria and green algae at frequencies of 20 kHz (1.47 W cm^{-3}), 582 kHz (1.32 W cm^{-3}), 862 kHz (0.41 W cm^{-3}) and 1144 kHz (1.015 W cm^{-3}). *M. aeruginosa*, *A. flos-aquae* and *S. suspicatus* with *Melosira* sp. were exposed to ultrasonic reactor. In this study, cell

growth of *M. aeruginosa*, *S. suspicatus* and *A. flos-aquae* and decreased by 16%, 20% and 99%, respectively. Also, at frequency of 20 kHz, cell count of *Melosira* sp. decreased by 83%. However at high frequencies of 582 kHz, 862 kHz and 1144 kHz, cell count of *Melosira* sp. decreased by 50%, 11% and 6%, respectively. This study demonstrated that cell wall and shape of filamentous cyanobacteria and green algae are causes for these differing degrees of susceptibility to ultrasonication. This study showed that at frequencies of 42 kHz (0.07 W cm^{-3}), 862 kHz (0.41 W cm^{-3}) and 200 kHz (0.015 W cm^{-3}), *Chlorophyceae* spp., *A. flos-aquae* and *M. aeruginosa* cells decreased to 100%, 99% and 95% after 130 s, 5–500 s and 30 s, respectively [46].

Effectiveness of sonochemical reactor (20 kHz, 200 kHz and 1.7 MHz) on removal of *S. platensis* were investigated by Hao et al. In this study, five samples of *S. platensis* cells were exposed to sonication for 5 min at 20 kHz and various power levels 20, 40, 60 and 80 W. In this experiment, cell growth of *S. platensis* decreased by percentages of 43%, 45%, 48% and 48%, respectively. After sonication at 20 W, there was a 26% decrease the final *S. platensis* cell count. The final cell count decreased by 40% at 40 W, 42% at 60 W and 44% at 80 W. These experiments showed that 40 W was the most efficient and economic power for sonication at 20 kHz [47,48].

Also, in this study, three samples of *S. platensis* cells were exposed for 5 min at 20 kHz, 200 kHz and 1.7 MHz at power of 40 W. The 5 min sonication duration at 20 kHz decreased the beginning of cell growth by 44% and the final cell count after 6 days by 21%. Also, at 1.7 MHz, the rates were 63% and 36%. But at 200 kHz the rates were 69% and 60%, respectively. This study, as well as other related research demonstrated that acoustic cavitation and collapsing phenomenon of the gas vesicles was the most efficient in controlling *S. platensis* cell growth [47,48].

The effects of ultrasonic waves on removal of *S. platensis* were investigated by Tang et al. Ultrasonic waves were tested at 1.7 MHz and low intensity (0.6 W cm^{-2}) to prevent growth of algae cells in water. The growth rate of *S. platensis* was reduced to 39% after 5 min. *S. platensis* was reduced from water by 30% and 60% with an exposure time of 12 min every 3 days and for 12 min every 11 days, respectively. It is suggested that sonication time, cell structure, chlorophyll *a* concentration and frequency are essential for effective inhibition of algal growth [49].

The effect of ultrasonic waves on removal of cyanobacterium cells such as *M. aeruginosa* (gas-vacuolate cyanobacterium) and *Synechococcus* spp. (gas-vacuole negative cyanobacterium) were studied by Tang et al. Sonoreactor (1.7 MHz, 5 min, 0.6 W cm^{-2}) were employed in this research. At 1.7 MHz, *M. aeruginosa* and *Synechococcus* spp. reduced 65% and 9% after 5 min sonication. In this study, was not influenced by ultrasonic reactor. Results showed that only *M. aeruginosa* cells were sensitive to ultrasonication. On the other hand, acoustic cavitation is responsible for controlling growth rate [50].

A study on the growth rate of *M. aeruginosa* (UTEX 2388) was investigated by Chi-Yong et al. In this study, a liquid culture of *M. aeruginosa* cells sonicated with ultrasonic processor that power and frequency were 630 W and 22 kHz respectively. The characteristics of ultrasonic processor were 50% duty, output 5%, 0.6 m in diameter and 0.7 m deep. Experiments showed that removal percentage of *M. aeruginosa* cells in the water samples was 13% after 3 day of sonication for 3 min in day. The disintegration process resulting from ultrasonication is the production of free radicals that damage *M. aeruginosa* cells [51].

Lee et al. suggested that ultrasonication at 100 W and wave frequency of 200 kHz resulted in the disruption and decay of cyanobacterial blooms (*Microcystis* spp.) gas vacuoles and sedimentation of cyanobacterial blooms. In another study, Lee et al. reported that the sonoreactor at 28 kHz and 700 W may decay the cyanobacteria gas vacuoles and deposit the *Microcystis* spp. in eutrophic lakes after treatment by ultrasonic waves. Sonicated *Microcystis* cells regenerated their collapsed gas vacuoles after 24 h and 36 h under aerated and non-aerated conditions, respectively. The rate that gas vacuoles regenerated was 87% after 60 h in non-aerated conditions. Sonicated cyanobacteria cells

under the process of aeration did not regenerate their gas vacuoles [52–54].

A study by Zhang et al. reported that a system at 0.32 W/mL and 25 kHz ultrasonic exposure for 5 min reduced amounts of *M. aeruginosa* cells by 11%, the photosynthetic activity by 40.5% and chlorophyll a concentration by 21.3% [55]. In another study, evaluated a range of ultrasonic generators at frequencies of 20, 80, 150, 410 and 1320 kHz by Zhang et al. This study showed that there was an increase in the *M. aeruginosa* removal rate constant from 150 kHz to 410 kHz [56].

Zhang et al. employed an ultrasonic pilot-scale reactor at 150 kHz and 30 W for growth inhibition of algal blooms. An increasing intensity from 23.6 Wcm⁻² to 47.2 Wcm⁻² increased the effectiveness of *M. aeruginosa* cell removal. In this study, the exposure time was 1–5 s and intensity was 47.2 Wcm⁻². This study showed that ultrasonication of an algae sample, resulted in the removal of chlorophyll a (5%–35%). In this experiment, 10% of *M. aeruginosa* cells were separated from solution by gravitational settling [57].

The effect of ultrasonication on reducing microcystins dissolved in *M. aeruginosa* suspensions were studied by Ma et al. In this study, a sonoreactor was operated at 20 kHz, 30 W, 60 W and 90 W for 5 min. Amounts of microcystins were reduced by 18%, 50% and 64%, respectively. These results showed that ultrasonic power was an effective parameter for removing microcystins dissolved in *M. aeruginosa* suspensions [58].

Wu et al. evaluated the effect of hydrodynamic cavitation on *M. aeruginosa* by an ultrasonic reactor at frequencies of 40 kHz (0.0466 Wcm⁻³) and 864 kHz (0.0929 Wcm⁻³). Results showed that after 30 min the concentration of *M. aeruginosa* decreased by 4% and 61%, respectively [59].

Wu et al. investigated the impact of ultrasonic at different frequencies on *M. aeruginosa*. This study showed that at frequencies of 20 kHz

(0.0403 Wcm⁻³), 580 kHz (0.0041 Wcm⁻³), 580 kHz (0.0216 Wcm⁻³), 1146 kHz (0.0018 Wcm⁻³) and 1146 kHz (0.0124 Wcm⁻³), *M. aeruginosa* cells decreased to 39.25%, 24.55%, 59.33%, 14.77% and 66.19% after 30 min, respectively. Low frequency of 20 kHz (0.0403 Wcm⁻³) also results in destroyed of cells [60].

Joyce et al. reported that using an ultrasonic reactor at a frequency of 580 kHz and intensities of 0.0018, 0.0210 and 0.0490 W cm⁻³ resulted in removing *M. aeruginosa* by 13%, 37% and 47.5%, respectively [61].

Effectiveness of ultrasonic irradiation (A polyethylene cylinder reactor, 10 cm in diameter) at a frequency of 200 kHz and input power of 3 W on removal of *M. aeruginosa* cells was studied by Spisukomwong et al. In this experiment, *M. aeruginosa* cells decreased in water treated after 30 s of sonication by 95%. A scanning electron microscope (SEM) picture of *M. aeruginosa* cells showed that ultrasonication (acoustic cavitation) effectively damaged cells of *M. aeruginosa* [62].

The impact of frequency at 20 kHz, exposure times (5, 10, 15, 20 min) and power intensities (0.043, 0.085, 0.139, 0.186 and 0.32 WmL⁻¹) on suspensions of *M. aeruginosa*, *A. circinalis* and *Chlorella* spp. were examined by Rajasekhar et al. For all power intensities and sonication times, the highest reduction rate in *M. aeruginosa* cell number was observed after 10 min of sonication. Also, in this study, the inhibition of *Chlorella* spp. (lack of gas vacuoles) growth rate was investigated. Experiments showed that under the same ultrasonication conditions, exposure to ultrasonic waves at 0.085 WmL⁻¹ had insufficient impact on the *chlorella* spp. For the cyanobacterial and algal cells tested in this study, under the same ultrasonication conditions, the order of decreasing growth inhibition at 0.085 WmL⁻¹ was: *A. circinalis* > *M. aeruginosa* > *Chlorella* spp. According to this research, the growth inhibition of the cyanobacterial and algal species cells was attributed mainly to sonication of gas vacuoles. Acoustic cavitation had a minimal impact on *Chlorella* spp. [63].

Table 1

Summary of the researches on the application of sonication conditions for cyanobacterial and algal cells control.

Cyanobacterial and algal cells	Frequency	Power or intensity/Density	Sonication time	Removal percentage	Ref.
<i>M. aeruginosa</i>	862 kHz	0.41 W cm ⁻³	5–500 s	1	46
<i>A. flos-aquae</i>	862 kHz	0.41 W cm ⁻³	5–500 s	99	
<i>S. suspicatus</i>	862 kHz	0.41 W cm ⁻³	5–500 s	20	
<i>Melosira</i> sp.	20 kHz	1.47 W cm ⁻³	5–500 s	83	
<i>Melosira</i> sp.	582 kHz	1.32 W cm ⁻³	5–500 s	50	
<i>Melosira</i> sp.	862 kHz	0.41 W cm ⁻³	5–500 s	11	
<i>Melosira</i> sp.	1144 kHz	1.015 W cm ⁻³	5–500 s	6	
<i>S. platensis</i>	20 kHz	20 W	5 min	43	47
	20 kHz	40 W	5 min	45	
	20 kHz	60 W	5 min	48	
	20 kHz	80 W	5 min	48	
<i>S. platensis</i>	200 kHz	40 W	5 min	60	48
	1.7 MHz	40 W	5 min	36	
<i>S. platensis</i>	1.7 MHz	0.6 W cm ⁻²	9 min	50	49
<i>M. aeruginosa</i>	1.7 MHz	0.6 W cm ⁻²	5 min	65	50
<i>Synechococcus</i> spp.	1.7 MHz	0.6 W cm ⁻²	5 min	9	
<i>M. aeruginosa</i>	22 kHz	630 W	3 min	13	51
<i>M. aeruginosa</i>	25 kHz	0.32 W ml ⁻¹	5 min	11	55
<i>M. aeruginosa</i>	150 kHz	47.2 W cm ⁻²	5 s	10	57
<i>M. aeruginosa</i> (Microcystin)	20 kHz	30, 60, 90 W	5 min	18,50,64	58
<i>M. aeruginosa</i>	40 kHz	0.0466 W cm ⁻³	30 min	4	59
	864 kHz	0.0929 W cm ⁻³	30 min	39.25	60
<i>M. aeruginosa</i>	20 kHz	0.0403 W cm ⁻³	30 min	24.55	
	580 kHz	0.0041 W cm ⁻³	30 min	59.33	
	580 kHz	0.0216 W cm ⁻³	30 min	14.77	
	1146 kHz	0.0018 W cm ⁻³	30 min	66.19	
	1146 kHz	0.0124 W cm ⁻³	30 min	39.25	
<i>M. aeruginosa</i>	580 kHz	0.018 W cm ⁻³	30 min	13	61
		0.0210 W cm ⁻³	30 min	37	
		0.0490 W cm ⁻³	30 min	47.5	
<i>M. aeruginosa</i>	200 kHz	3 W	30 s	95	62
<i>M. aeruginosa</i>	20 kHz	0.043, 0.085, 0.139, 0.186,	5, 10, 15,	41.15–68.25	63
<i>A. circinalis</i>		0.32 W cm ⁻³	20 min		
<i>Chlorella</i> spp					
Algal bloom and chlorophyll a	40 kHz	60 W	15 s	13 and 11	64
<i>Chlorophyceae</i> spp.	42 kHz	0.07 W cm ⁻³	130 s	100	65

Heng et al. showed that using ultrasonic reactor waves mainly led to increased aggregation and sedimentation of algal bloom in the Luan River. The water samples were exposed to the ultrasonic wave at frequency of 40 kHz and ultrasound power level of 60 W and the sonication time of 15 s. In this study, algae cells and chlorophyll *a* removed in water sample after 15 s of sonication by 13% and 11%, respectively [64].

The application of ultrasound reactor to sonolysis of *Chlorophyceae* spp. was evaluated in laboratory conditions by Dehghani and Changani. This study, showed that short exposure to ultrasonic waves can result in a loss of buoyancy so that by 15, 30, 45, 60, 75, 90, 110 and 130 s of sonolysis (42 kHz) about 7%, 11%, 34%, 51%, 64%, 81%, 95.5% and 100% of the *Chlorophyceae* spp. present are destroyed respectively. The results of using ultrasound reactor at 42 kHz showed that sonolysis of *Chlorophyceae* spp. occurred rapidly. It is concluded that with this frequency 100% of the *Chlorophyceae* spp. can be destroyed in 130 s [25]. Another study showed that short exposure to ultrasonic waves collapsed *Chlorophyceae* spp. gas vacuoles resulting in a loss of buoyancy after sonication [65].

The effectiveness of the ultrasound at frequencies of 20, 580, 864, 1146 kHz on degradation of both *C. Concordia* and *D. salina* in water investigated by Yamamoto et al. This study showed that high frequency is more effective than low-frequency for the degradation of algal cells (20 < 580 < 864 < 1146 kHz). The results showed that suitable sonication for degradation of algal cells dependent on the mechanical properties of cells [66].

Table 1 presented a summary of the researches on the application of sonication conditions for cyanobacteria control.

4. Sonochemical degradation of cyanotoxins from water



Cyanobacteria are organisms that are present in many aquatic ecosystems including freshwater and marine. Cyanobacterial blooms toxin released by blue-green algae are the most toxin that found in drinking water supply and aquatic environment, as well as responsible for poisoning fish, shrimp, crayfish and have adverse effects for human health including liver damage, kidney damage, tumor promotion, gastrointestinal etc. [67–71]. The World Health Organization reported a drinking water guideline of 1 µg/L for microcystin-LR [72].

The cyanotoxins described by the literature are categorized as follows, based on the health impacts [73–79] [Table 2].

Microcystins (Fig. 1) [80] are very stable and conventional water treatment processes unable for removal of them. In recent years, sonoreactors as an oxidation technology has been used to inactivation of algal bloom and toxins. Several papers have evaluated the persistence of cyanobacterial toxins in the water ecosystem [25,77,78,81].

In one study, Song et al. observed that sonication at a frequency of 640 kHz resulted in microcystin-LR degradation. Experiments showed that after 3 min and 6 min, the concentration of microcystin-LR is reduced to 50% and 80%, respectively. This study showed that hydroxyl radical produced from ultrasonic waves is responsible for removal of toxin [82–84].

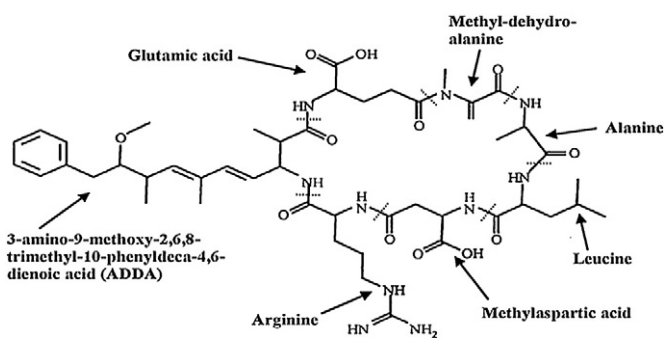


Fig. 1. Molecular structure of microcystin-LR (Sielaff et al. [80]).

Hudder et al. observed that at a frequency of 640 kHz (intensity 0.5 Wcm^{−3}) and sonication time of 1.5 h resulted within 99% of microcystin-LR degradation [85]. Rajasekhar et al. showed that power intensities (0.32 Wcm^{−3} and 0.043 Wcm^{−3}) and sonication time for >10 min, led to increase of microcystin toxin concentration in the sonicated sample [63].

Sonodegradation of microcystin toxin produced by *M. aeruginosa* in water were studied by Ma et al. This study showed that sonication after 5 min would not induce the increase of microcystin in water. After 7 min sonication at 20 kHz and 30 W, it began to increase in water. In these experiments, samples were contacted to 5 min sonication at 20 kHz with power of 60 W and 90 W. After 5 min sonication the microcystin concentration were decreased by 12% and 4%, respectively. In this research, microcystin solution was exposed to sonoreactor at 20 kHz with power of 30 W and 30 min ultrasonication. After 30 min sonication, the microcystin concentrations were reduced by 65%. The results showed that sonication was an effective technology for removing microcystin toxin in water. In this study, three microcystin concentration were exposed at 20 kHz with powers of 30 W, 60 W and 90 W. The microcystin sample was examined at 1 min, 5 min, 10 min and 20 min. In these experiments, after 5 min sonication with powers of 30 W, 60 W and 90 W, the microcystin were reduced by 18%, 50% and 64%, respectively. Degradation rates of microcystins samples with power of 60 W and 90 W became slow after 5 min ultrasonication. Also, four microcystin samples were exposed to sonicator at 20 kHz, 150 kHz, 410 kHz and 1.7 MHz with power of 30 W. The microcystin samples were examined at 1 min, 5 min, 10 min and 20 min. Results showed that, after 20 min sonication at 20 kHz, 150 kHz, 410 kHz and 1.7 MHz, with power at 30 W, the microcystin concentration were reduced by 58%, 71%, 65% and 54%, respectively [58].

Ahn et al. - demonstrated that at a frequency of 22 kHz (intensity 0.63 W cm^{−3}) leads to degradation of microcystin concentration from 66% to 0.3% after 3 day ultrasonication [1].

The effect of ultrasonic power on degradation of *M. aeruginosa* investigated by Zhang et al. They reported that 80 W increased microcystin concentrations in water sample [55]. Lee et al. (2001) demonstrated

Table 2
Characteristics of cyanotoxins [73–79].

cyanotoxins	formula	Classified	Chemical Structure	Example	Health effects
Anatoxin-a	C ₁₀ H ₁₅ NO	Neurotoxin	bicyclic amine alkaloid	Anabaena, Oscillatoria, Aphanizomenon	paralysis and death
Anatoxin-a(s)	C ₇ H ₁₇ N ₄ O ₄ P	Neurotoxin	Alkaloid	Anabaena	destruction of muscle function
Saxitoxins	C ₁₀ H ₁₇ N ₇ O ₄	Neurotoxin	Alkaloid	Anabaena, Aphanizomenon, Cyndrospermopsis	blocks sodium channels of nerve cells
Microcystins - LR	C ₄₉ H ₇₄ N ₁₀ O ₁₂	Hepatotoxin	Monocyclic heptapeptide	Microcystis, Anabaena, Hapalosiphon, Nostoc, Anabaenopsis, Oscillatoria	Liver damage
Nodularins	C ₄₁ H ₆₀ N ₈ O ₁₀	Hepatotoxin	Pentapeptide	Nodularia	Tumor promoter
Cylindrospermopsin	C ₁₅ H ₂₁ N ₅ O ₇ S	Cytotoxin	Alkaloid	Cylindrospermopsis, Aphanizomenon	Liver damage
					Tumor promoter
					Kidney damage

that after 5 min sonication (120 W and 28 kHz) the microcystin toxin in suspension of blue green algae did not increase [53].

Zhang et al. reported that at 0.32 W cm⁻³ and 25 kHz sonication for 5 min, reduced the microcystin by 7%, but after 14 days culturing, toxin concentrations of sonicated solution were 14% and 16%, respectively [56].

5. Conclusions



Sonochemical reactor as a novel technology for removal of cyanobacterial and algal cells has been under evaluation for the past decade. The impact of ultrasonication on cyanobacterial and algal cells growth are by the damage of gas vacuoles and cell wall, and reduction of photosynthetic activities. Effect of ultrasonication on the growth rate of algal cells and cyanobacteria is mostly dependent on the length of sonication time, its frequency and power. At high frequency (1.7 MHz) and high power free radicals are produced from the ultrasonication of water sample, also damage cell structure and chlorophyll *a* of cells. At low frequency (20 kHz) and low power also results in destroyed of cells. Also, these studies show that hydroxyl radicals are very important for the degradation of toxin after sonication.

References

- [1] C.Y. Ahn, M.H. Park, S.H. Jeong, H.S. Kim, K.Y. Jang, H.M. Oh, Growth inhibition of cyanobacteria by ultrasonic radiation: laboratory and enclosure studies, *Environ. Sci. Technol.* 37 (2003) 3031.
- [2] C.Y. Ahn, S.H. Jeong, A. Choi, H.S. Kim, K.Y. Jang, H.M. Oh, Selective control of cyanobacteria in eutrophic pond by a combined device of ultrasonication and water pumps, *Environ. Technol.* 28 (2007) 371.
- [3] S. Al-Hamdani, C. Burnett, G. Durrant, Effect of low-dose ultrasonic treatment on *Spirulina maxima*, *Aquac. Eng.* 19 (1998) 17.
- [4] Y. Asakura, M. Maebayashi, S. Koda, Study on efficiency and characterization in a cylindrical sonochemical reactor, *Chem. Eng. Jpn.* 38 (2005) 1008.
- [5] S.R. Auzay, J.B.E. Naffrechoux, Comparison of characterization methods in high frequency sonochemical reactors of differing configurations, *Ultrason. Sonochem.* 17 (2010) 547.
- [6] A. Balasubrahmanyam, A.B. Pandit, Experimental investigation of cavitation bubble dynamics under multi-frequency system, *Ultrason. Sonochem.* 15 (2008) 578.
- [7] M.H. Dehghani, A.A. Najafpoor, K. Azam, Using sonochemical reactor for degradation of LAS from effluent of wastewater treatment plant, *Desalination* 250 (2010) 82.
- [8] M.H. Dehghani, G.H.R. Jahed, F. Vaezi, Evaluation of USR technology on the destruction of HPC organisms, *Pak. J. Biol. Sci.* 9 (2006) 2127.
- [9] M.H. Dehghani, Effectiveness of ultrasound on the destruction of *E. coli*, *Am. J. Environ. Sci.* 1 (2005) 187.
- [10] M.H. Dehghani, A.M. Fadaei, Sonochemical kinetic model of diazinon and malathion pesticides degradation in aqueous solution, *Indian J. Sci. Technol.* 6 (2013) 3876.
- [11] A.M. Fadaei, M.H. Dehghani, A. Rahimi Foroshan, M. Sadeghi, Using Sonophotodegradation Technology for Removal of organophosphorus pesticides in aqueous solution, *Asian J. Chem.* 25 (2013) 7517.
- [12] S. Broekman, O. Pohlmann, E.S. Beardwood, E.C. Meulenaar, Ultrasonic treatment for microbiological control of water systems, *Ultrason. Sonochem.* 17 (2010) 1041.
- [13] P.R. Gogate, V.S. Sutkar, A.B. Pandit, Sonochemical reactors: important design and scale up considerations with a special emphasis on heterogeneous systems, *Chem. Eng.* 166 (2011) 1066.
- [14] P.R. Gogate, Cavitation reactors for process intensification of chemical processing applications: a critical review, *Chem. Eng. Prof.* 47 (2008) 515.
- [15] Y. Kojima, Y. Asakura, G. Sugiyama, S. Koda, The effects of acoustic flow and mechanical flow on the sonochemical efficiency in a rectangular sonochemical reactor, *Ultrason. Sonochem.* 17 (2009) 978.
- [16] E. Joyce, The development and evaluation of ultrasound for the treatment of bacterial suspension, *Ultrason. Sonochem.* 10 (2002) 315.
- [17] F. Laugier, C. Andriantsiferana, A.M. Wilhelm, H. Delmas, Ultrasound in gas-liquid systems: effects on solubility and mass transfer, *Ultrason. Sonochem.* 15 (2008) 965.
- [18] E. Kim, M. Cui, M. Jang, B. Park, Y. Son, J. Khim, Investigation of sonochemical activities at a frequency of 334 kHz: the effect of geometric parameters of sonoreactor, *Ultrason. Sonochem.* 21 (2014) 1504.
- [19] E.V. Rokhina, P. Lens, J. Virkutyte, Low-frequency ultrasound in biotechnology: state of the art, *Trends Biotechnol.* 27 (2009) 298.
- [20] M. Shayeghi, M.H. Dehghani, A.M. Fadaei, Removal of malathion insecticide from water by employing acoustical wave technology, *Iran. J. Public Health* 40 (2011) 122.
- [21] A. Rodriguez-Molares, S. Dickson, P. Hobson, C. Howard, A. Zander, M. Burch, Quantification of the ultrasound induced sedimentation of *Microcystis aeruginosa*, *Ultrason. Sonochem.* 21 (2014) 1299.
- [22] Y. Son, M. Lim, J. Khim, Investigation of acoustic cavitation energy in a large scale sonoreactor, *Ultrason. Sonochem.* 16 (2009) 552.
- [23] V.S. Sutkar, P.R. Gogate, Design aspects of sonochemical reactors: techniques for understanding cavitation activity distribution and effect of operating parameters, *Chem. Eng.* 155 (2009) 26.
- [24] V.S. Sutkar, P.R. Gogate, Mapping of cavitation activity in high frequency sonochemical reactor, *Chem. Eng.* 158 (2010) 296.
- [25] M.H. Dehghani, F. Changani, The effect of acoustic cavitation on chlorophyll *a* from effluent of wastewater treatment plant, *Environ. Technol.* 27 (2006) 963.
- [26] N. Scheifhacken, H. Horn, L. Paul, Comparing *in situ* particle monitoring to microscopic counts of plankton in a drinking water reservoir, *Water Res.* 44 (2010) 3496.
- [27] Z. Romanowska-Duda, J. Mankiewicz, M. Tar-Czynska, Z. Walter, M. Zalewski, The effect of toxic cyanobacteria (blue green algae) on water plants and animal cells, *Pol. J. Environ. Stud.* 11 (2002) 561.
- [28] J.A. Westrick, Cyanobacterial toxin removal in drinking water treatment processes and recreational waters chapter 13, in: H.K. Hudnell (Ed.), *Cyanobacterial Harmful Algal Blooms. State of the Science and Research Needs*, Springer 2008, pp. 275–290.
- [29] T. Jurczak, M. Tarczynska, K. Izdorzyczyk, J. Mankiewicz, M. Zalewski, J. Meriluoto, Elimination of microcystins by water treatment processes examples from Sulejow reservoir, Poland, *Water Res.* 39 (2005) 2394.
- [30] M. Piontek, R. Piontek, Hazards to humans and animals caused by the presence of cyanobacteria in aquatic ecosystems, *Natura* 8 (2003) 79.
- [31] A.E. Walsby, Gas vesicles, *Microbiological Reviews* 58 (1994) 144.
- [32] A. Kozak, Seasonal changes occurring over four years in a Reservoir's phytoplankton composition, *Pol. J. Environ. Stud.* 14 (2005) 4451.
- [33] M. Piontek, W. Czyżewska, Efficiency of drinking water treatment processes. Removing of phytoplankton with special consideration to cyanobacteria and improving physical and chemical parameters, *Pol. J. Environ. Stud.* 6 (1997) 2012.
- [34] A. Napiorkowska-Krzebietke, K. Stawiecki, J.P. Pyka, J. Hutorowicz, B. Zdanowski, Phytoplankton in relation to water quality of a mesotrophic lake, *Pol. J. Environ. Stud.* 22 (2013) 793.
- [35] T. Sobczynski, T. Joniak, The variability and stability of water chemistry in a deep temperate lake: results of long-term study of eutrophication, *Pol. J. Environ. Stud.* 22 (2013) 227.
- [36] N. Jakubowska, P. Zagajewski, R. Goldyn, Water blooms and cyanobacterial toxins in lakes, *Pol. J. Environ. Stud.* 22 (2013) 1077.
- [37] J.J. Chen, H.H. Yeh, The mechanisms of potassium permanganate on algae removal, *Water Res.* 39 (2005) 4420.
- [38] Y. Asakura, K. Yasuda, D. Kato, Y. Kojima, S. Koda, Development of a large sonochemical reactor at a high frequency, *Chem. Eng.* 139 (2008) 339.
- [39] Y. Chen, J. Liu, J. Ju, Flotation removal of algae from water, *Colloids Surf. B: Biointerfaces* 12 (1998) 49.
- [40] J. Ma, W. Liu, Effectiveness and mechanism of potassium ferrate (VI) preoxidation for algae removal by coagulation, *Water Res.* 36 (2002) 871.
- [41] J.D. Plummer, J.K. Edzwald, Effect of chlorine and ozone on algal cell properties and removal of algae by coagulation, *J. Water Supply Res. Technol. AQUA* 51 (2002) 307.
- [42] K. Nakano, T. Lee, M. Matsumara, In situ algae bloom control by the integration of ultrasonic radiation and jet circulation to flushing, *Environ. Sci. Technol.* 35 (2001) 941.
- [43] D.A. Franko, S.R. Taylor, B.J. Thomas, D. McIntosh, Effect of low-dose ultrasonic treatment on physiological variables in *Anabaena flos-aquae* and *Selenastrum capricornutum*, *Biotechnol. Lett.* 12 (1990) 219.
- [44] D.A. Franko, S. Al-Hamdani, G.-J. Joo, Enhancement of nitrogen fixation in *Anabaena flos-aquae* (cyanobacteria) via low-dose ultrasonic treatment, *J. Appl. Phycol.* 6 (1994) 455.
- [45] B.J. Thomas, D. McIntosh, S.R. Taylor, D.A. Franko, J. Ownby, Effect of low-dose ultrasonic treatment on growth rates and biomass yield of *Anabaena flos-aquae* and *Selenastrum capricornutum*, *Biotechnol. Tech.* 3 (1989) 389.
- [46] D. Purcell, Control of Algal Growth in Reservoirs With Ultrasound PhD. thesis School of Applied Sciences, Cranfield University, Centre for Water Science, Department of Sustainable Systems, UK, December 2009.
- [47] H. Hao, W. Wu, Y. Chen, J. Tang, Q. Wu, Cyanobacteria bloom control by ultrasonic irradiation at 20 kHz and 1.7 MHz, *J. Environ. Sci. Health A* 39 (2004) 1435.
- [48] H. Hao, M. Wu, Y. Chen, J. Tang, Q. Wu, Cavitation mechanism in cyanobacterial growth inhibition by ultrasonic irradiation, *Colloids Surf. B: Biointerfaces* 33 (2004) 151.
- [49] J. Tang, Q. Wu, H. Hao, Y. Chen, M. Wu, Growth inhibition of the cyanobacterium *Spirulina (Arthrospira) platensis* by 1.7 MHz ultrasonic irradiation, *J. Appl. Phycol.* 15 (2003) 37.
- [50] J.W. Tang, Q.Y. Wu, H.W. Hao, Y.F. Chen, M.S. Wu, Effect of 1.7 MHz ultrasound on a gas-vacuolate cyanobacterium and a gas-vacuole negative cyanobacterium, *Colloids Surf. B: Biointerfaces* 36 (2004) 115.
- [51] Y.A. Chi, H.P. Myung, H.J. Seung, S.K. Hee, Y.J. Kam, M.O. Hee, Growth inhibition of cyanobacteria by ultrasonic radiation: laboratory and enclosure studies, *Environ. Sci. Technol.* 37 (2003) 3031.
- [52] T. Lee, K. Nakano, M. Matsumara, A novel strategy for algae bloom control by ultrasonic irradiation, *Water Sci. Technol.* 46 (2002) 207.
- [53] T.J. Lee, K. Nakano, M. Matsumara, Ultrasonic irradiation for blue-green algae bloom control, *Environ. Technol.* 22 (2000) 383.
- [54] T.J. Lee, K. Nakano, M. Matsumara, A new method for the rapid evaluation of gas vacuoles regeneration and viability of cyanobacteria by flow cytometry, *Biotechnol. Lett.* 22 (2000) 1833.
- [55] G. Zhang, P. Zhang, H. Liu, B. Wang, Ultrasonic damages on cyanobacterial photosynthesis, *Ultrason. Sonochem.* 13 (2006) 501.
- [56] G. Zhang, P. Zhang, B. Wang, H. Liu, Ultrasonic frequency effects on the removal of *Microcystis aeruginosa*, *Ultrason. Sonochem.* 13 (2006) 446.
- [57] G. Zhang, P. Zhang, M. Fan, Ultrasound-enhanced coagulation for *Microcystis Aeruginosa* removal, *Ultrason. Sonochem.* 16 (2009) 334.

- [58] B.Z. Ma, Y.F. Chen, H.W. Hao, M.S. Wu, B. Wang, H.G. Lv, G.M. Zhang, Influence of ultrasonic field on microcystins produced by bloom-forming algae, *Colloids Surf. B: Biointerfaces* 41 (2005) 197.
- [59] X. Wu, The effect of ultrasound effect on algae in water, Faculty of Health and Life Sciences, Coventry University, Coventry, 2011 Ph.D. Thesis.
- [60] X. Wu, E.M. Joyce, T.J. Mason, Evaluation of the mechanisms of the effect of ultrasound on *Microcystis aeruginosa* at different ultrasonic frequencies, *Water Res.* 46 (2012) 2851.
- [61] E.M. Joyce, X. Wu, T.J. Mason, Effect of ultrasonic frequency and power on algae suspensions, *J. Environ. Sci. Health A* 45 (2010) 863.
- [62] P. Spisukomwong, N. Whangchai, Y. Yagita, K. Yuwadee, P. Nakaonmura, Effects of ultrasonic irradiation on degradation of microcystin in fish ponds, *Int. J. Agric. Biol.* 13 (2011) 67.
- [63] P. Rajasekhar, L. Fan, T. Nguyen, F.A. Roddick, Impact of sonication at 20 kHz on *Microcystis aeruginosa*, *Anabaena circinalis* and *Chlorella* sp., *Water Res.* 46 (2012) 1473.
- [64] L. Heng, N. Jun, H. Wen-jie, L. Guibai, Algae removal by ultrasonic irradiation-coagulation, *Desalination* 239 (2009) 191.
- [65] A.H. Mahvi, M.H. Dehghani, Evaluation of ultrasonic technology in removal of algae from surface waters, *Pak. J. Biol. Sci.* 8 (2005) 1457.
- [66] K. Yamamoto, P.M. King, X. Wu, T.J. Mason, E.M. Joyce, Effect of ultrasonic frequency and power on the disruption of algal cells, *Ultrason. Sonochem.* 24 (2015) 165–171.
- [67] I.R. Falconer, An overview of problems caused by toxic blue-green algae (cyanobacteria) in drinking water, *Environ. Toxicol.* 14 (1999) 5–12.
- [68] G.A. Codd, et al., Cyanobacterial toxins: risk management for health protection, *Toxicol. Appl. Pharmacol.* 203 (2005) 264–272.
- [69] L.E. Fleming, et al., Blue green algal (cyanobacterial) toxins, surface drinking water, and liver cancer in Florida, *Harmful Algae* 1 (2002) 157–168.
- [70] I.R. Falconer, A.R. Humpage, Cyanobacterial (blue-green algal) toxins in water supplies: Cylinderspermopsins, *Environ. Toxicol.* 21 (2006) 299–304.
- [71] A. Fristachi, J.L. Sinclair, Occurrence of cyanobacterial harmful algae blooms workgroup report, in: H.K. Hudnell (Ed.), *Cyanobacterial Harmful Algal Blooms: State of the Science and Research Needs*, Springer Science, New York, 2008.
- [72] WHO, Cyanobacterial Toxins: Microcystin-LR, Guidelines for Drinking Water Quality, World Health Organization, Geneva, 1998.
- [73] R.S. Yoo, et al., Cyanobacterial (Blue-Green Algal) Toxins: A Resource Guide, AWWA Research Foundation and AWWA, Denver, 1995.
- [74] K. Sivonen, G. Jones, Cyanobacterial Toxins, in: I. Chorus, J. Bartram (Eds.), *Toxic Cyanobacteria in Water: A Guide to their Public Health Consequences, Monitoring, and Management*, Routledge, London, 1999.
- [75] J. Bartram, et al., Introduction, in: I. Chorus, J. Bartram (Eds.), *Toxic Cyanobacteria in Water: A Guide to their Public Health Consequences, Monitoring, and Management*, Routledge, London, 1999.
- [76] T. Kuiper-Goodman, et al., Human health aspects, in: I. Chorus, J. Bartram (Eds.), *Toxic Cyanobacteria in Water: A Guide to their Public Health Consequences, Monitoring, and Management*, Routledge, London, 1999.
- [77] G. Newcombe, et al., Treatment options for microcystin toxins: similarities and differences between variants, *Environ. Technol.* 24 (3) (2003) 299–308.
- [78] J.D. Brookes, et al., Reservoir Management Strategies for Control and Degradation of Algal Toxins, AWWA Research Foundation, Denver, 2008.
- [79] World Health Organization, Guidelines for Drinking-Water Quality, Third edition, 2008 Geneva.
- [80] H. Sielaff, E. Dittmann, N. Tandeau de Marsac, C. Bouchier, H. Von Dohren, T. Brner, T. Schwecke, The *mcyF* gene of the microcystin biosynthetic gene cluster from *Microcystis aeruginosa* encodes an aspartate racemase, *Biochemistry* 373 (2003) 909–916.
- [81] J. Rapala, K. Lahti, K. Sivonen, S.J. Niemela, Biodegradability and adsorption on lake sediments of cyanobacterial hepatotoxins and anatoxin-a, *Lett. Appl. Microbiol.* 19 (1994) 423–428.
- [82] W. Song, T. Teshiba, K. Rein, K.E. O'Shea, Ultrasonically induced degradation and detoxification of microcystin-LR (cyanobacterial toxin), *Environ. Sci. Technol.* 39 (2005) 6300–6305.
- [83] W. Song, A.A. De La Cruz, K. Rein, K.E. O'Shea, Ultrasonically induced degradation of microcystin-LR and -RR: identification of products, effect of pH, formation and destruction of peroxides, *Environ. Sci. Technol.* 40 (2006) 3941–3946.
- [84] W. Song, S. Bardowell, K.E. O'Shea, Mechanistic study and the influence of oxygen on the photosensitized transformations of microcystins (cyanotoxins), *Environ. Sci. Technol.* 41 (2007) 5336–5341.
- [85] A. Hudder, W. Song, K.E. O'Shea, P.J. Walsh, Toxicogenomic evaluation of microcystin-LR treated with ultrasonic irradiation, *Toxicol. Appl. Pharmacol.* 220 (2007) 357–364.